

ulms-highgrade-pulm-mets-er-pr-neg-q qvt

Selected general biomarker report — biomarker screening coverage and gaps

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Selected general biomarker report

Which biomarkers on the selected panel this patient has been tested for, and which have never been measured. The panel covers targets established in this tumor type, targets relevant across a subset of tumor types, and biomarkers that predict benefit regardless of where the tumor started. An unmeasured biomarker below is a gap in the record, not a prediction that the patient carries it.

Surveyed: 15 biomarkers. **No usable result on file:** 15. **Measured but not decision-grade:** 0. **Already established:** 0.

Every one of the fifteen biomarkers surveyed here is a gap, and none is measured to decision grade. No sequencing exists from either specimen; the only stains ever run are ER and PR. Twelve were never measured; three, PRAME, MAGE-A4 and NY-ESO-1, are unassessable rather than negative: no HLA class I type is on file to score them against. Four gaps are essential, all order-now: NTRK fusion, MSI/dMMR (four stains on an existing block, plus Lynch implications for a woman in her thirties), TMB, and HRD/BRCA, the entry this histology most earns: olaparib plus temozolomide gave a 27% response rate and 6.9-month median PFS in pretreated uLMS.

The list collapses to three orders and no new biopsy; tissue exists from the 2024 hysterectomy and the 2026 lung core. Comprehensive DNA plus RNA profiling on the 2024 resection is the highest-value order: it closes NTRK, TMB and HRD at once and returns TP53, RB1, ATRX and MED12 without a separate test. It must include RNA. NTRK breakpoints sit in introns that DNA-only panels miss, and this spindle-cell tumor was already misread once as benign. Second, one archival block cut for MMR, PD-L1, HER2 and B7-H3 IHC, with PD-L1 read on the lung core. Third, a blood draw for HLA typing and the germline HR panel. All three can run beside the planned LMS-04 induction without delaying it, and since the tumor-agnostic indications require prior therapy anyway, the results land exactly when they can be acted on: at progression.

Two absences are deliberate. ER and PR <1% are confirmed decision-grade negatives with no survey row because neither reference panel lists them; the validation track is checking which specimen was stained, since at that cutoff a failed stain on the 2024 block reads the same as a true negative. The other 66 of 72 protein targets were set aside as epithelial markers a mesenchymal tumor does not plausibly express. B7-H3 stayed, with the best odds of a positive here: 97% of soft tissue sarcomas stain positive, roughly 69% high, though a positive buys trial screening, not a prescription. Most of what remains will be negative if tested. The argument for testing is the size of the option a positive opens, not the odds of one.

Gaps: biomarkers with no usable result on file

Microsatellite instability / mismatch-repair deficiency (MSI-H / dMMR)

Priority: essential **Scope:** Tumor-agnostic **Status:** Not measured

MSI-H and dMMR are read independently of primary site. Nothing in this record speaks to mismatch repair, and she becomes a checkpoint-blockade candidate the moment first-line chemotherapy stops working.

Mismatch-repair status has never been assessed. Uterine leiomyosarcoma is nearly always proficient, so a negative is the likely result, and the case for testing is that four stains on an existing block close a tumor-agnostic PD-1 indication that would otherwise linger as an open question at progression. Loss of any one protein would

also carry Lynch implications for a woman in her thirties and for her first-degree relatives.

Recommended assay. MMR protein IHC (MLH1, PMS2, MSH2, MSH6) (IHC · Archival FFPE block; either specimen works, the 2024 resection has more tissue to spare · 3-5 days)

Bundles with. Single archival block cut for the IHC set (MMR, PD-L1, HER2, B7-H3)

A positive result would open.

- pembrolizumab (PD-1 inhibitor · tumor-agnostic approval · NCT02628067)
- dostarlimab (PD-1 inhibitor · tumor-agnostic approval · NCT02715284)

Next step. Order now

References. nct:NCT02628067; nct:NCT02715284

NTRK1 / NTRK2 / NTRK3 gene fusion

Priority: essential **Scope:** Tumor-agnostic **Status:** Not measured

Primary site does not enter the TRK indications: any solid tumor carrying an in-frame NTRK fusion qualifies. Spindle-cell morphology is also the setting in which an NTRK-rearranged mesenchymal neoplasm is most often signed out as something else, and this tumor was already signed out as a benign leiomyoma once.

NTRK fusion is untested and nothing currently planned would find it. Prevalence in uterine leiomyosarcoma is very low, so a negative is the expected result, but a positive converts a chemotherapy-only landscape into an oral TRK inhibitor with high and durable response rates, and the assay rides along on the same RNA fusion panel that answers RET. A DNA-only panel does not close this question: NTRK breakpoints sit in large introns and are missed without RNA.

Recommended assay. RNA-based fusion NGS panel covering NTRK1/2/3 (NGS_panel · Archival FFPE. The 2024 hysterectomy specimen is preferred for volume; the 2026 lung core is acceptable and reflects current disease · 2-3 weeks)

Bundles with. Comprehensive DNA plus RNA genomic profiling on archival tissue, which also returns TMB, MSI, RET and BRAF, the HR genes, and the TP53 / RB1 / ATRX / MED12 backbone of this histology

A positive result would open.

- larotrectinib (TRK inhibitor · tumor-agnostic approval · NCT02576431)
- entrectinib (TRK / ROS1 / ALK inhibitor · tumor-agnostic approval · NCT02568267)
- repotrectinib (next-generation TRK inhibitor · tumor-agnostic approval · NCT03093116)

Next step. Order now

References. nct:NCT02576431; nct:NCT02568267; nct:NCT03093116

Tumor mutational burden, high (TMB-H)

Priority: essential **Scope:** Tumor-agnostic **Status:** Not measured

TMB at or above 10 mutations per megabase supports pembrolizumab whatever the primary site. No sequencing of any kind has been performed here, so TMB is unknown rather than low.

TMB is unmeasured because no sequencing has been done at all. Leiomyosarcoma is typically copy-number driven and low-TMB, so a high result would be unusual, but the panel that reports it also reports MSI, the HR genes, the fusions and the TP53 / RB1 / ATRX / MED12 alterations that define this histology, which makes TMB close to free once the order is placed. Specify a validated comprehensive assay: a TMB figure from a small hotspot panel does not satisfy the indication.

Recommended assay. Validated comprehensive genomic profiling panel reporting TMB in mutations per megabase (NGS_panel · Archival FFPE, 2024 uterine resection preferred for tumor content and volume · 2-3 weeks)

Bundles with. Comprehensive DNA plus RNA genomic profiling on archival tissue (the single order that also carries MSI, NTRK, RET, BRAF and the HR genes)

A positive result would open.

- pembrolizumab (PD-1 inhibitor · tumor-agnostic approval · NCT02628067)

Next step. Order now

References. nct:NCT02628067

Homologous recombination deficiency / BRCA1 / BRCA2 pathogenic variant

Priority: essential **Scope:** Relevant across a tumor subset **Status:** Not measured

Uterine leiomyosarcoma has a recognised homologous-recombination-deficient subset, with BRCA2 deletions and mutations reported, and NCI protocol 10250 showed olaparib plus temozolomide is active in pretreated disease. Of the nine entries on this panel, this is the one uterine leiomyosarcoma most earns.

Homologous-recombination status is untested, and roughly a tenth of uterine leiomyosarcomas carry an HR pathway mutation with a larger fraction functionally deficient by RAD51 assay. A pathogenic BRCA2 or other HR gene finding would put olaparib plus temozolomide and PARP-inhibitor trials on the table for later lines, and it would trigger germline testing that matters both for her and for relatives at this age. Somatic-only testing leaves the germline question open, so order the blood panel alongside rather than after.

Recommended assay. Tumor NGS covering BRCA1/2, PALB2, RAD51C, RAD51D and ATM with a genomic-instability (HRD) score, plus a germline multigene panel on blood (NGS_panel · Archival FFPE for the somatic panel; whole blood for the germline panel · 2-3 weeks tumor, 2-4 weeks germline)

Bundles with. Comprehensive DNA plus RNA genomic profiling on archival tissue; germline panel on the same blood draw as HLA typing

A positive result would open.

- olaparib plus temozolomide (PARP inhibitor plus alkylating agent · trial only · NCT03880019)
- PARP inhibitors (PARP inhibitor · approved, other tumor type)

Next step. Order now

References. pmid:37467452; pmid:37143137; nct:NCT03880019

B7-H3 (CD276) surface protein expression

Priority: high **Scope:** Relevant across a tumor subset **Status:** Not measured

B7-H3 sits on tumor cells and tumor vasculature across soft tissue sarcoma: 97% of specimens positive and about 69% high in a 153-patient series covering 15 subtypes, leiomyosarcoma among them. Of the surface targets on the panel, this is the one most likely to be positive in this histology.

B7-H3 has not been stained, and it carries the best odds of a positive result of any surface target here. Nothing is approved against it, so a positive buys trial access rather than a prescription, and the ADC, CAR-T and bispecific programs generally require documented expression at screening. Staining it on the same block as the other IHC keeps a later trial referral from stalling while tissue is hunted down.

Recommended assay. B7-H3 (CD276) IHC on tumor tissue (IHC · Archival FFPE; the 2026 lung core reflects the treated-disease phenotype, the 2024 resection has more tissue · 1 week)

Bundles with. Single archival block cut for the IHC set (MMR, PD-L1, HER2, B7-H3)

A positive result would open.

- ifinatamab deruxtecan (ADC · trial only)
- enoblituzumab (mAb · trial only)
- vobramitamab duocarmazine (ADC · trial only)

Next step. Bundle with planned testing

References. pmid:39478506

HLA class I genotype

Priority: high **Scope:** Relevant across a tumor subset **Status:** Not measured

Every intracellular antigen in the selected panel that is plausible here (PRAME, MAGE-A4, NY-ESO-1) is reachable only through a restricting HLA class I allele, most often HLA-A*02:01. Without a genotype those three rows sit as unassessable rather than negative.

HLA class I type is not on file, which leaves three antigen rows in this survey unassessable rather than negative. It is a blood draw, it costs little, it spends no tumor tissue, and roughly half of patients of European ancestry carry HLA-A*02:01, so the result either opens the TCR-T and ImmTAC class for later consideration or closes it cleanly. Ordering it now avoids a two-step delay if she progresses through first-line chemotherapy and trial options come into play.

Recommended assay. HLA class I genotyping at 4-digit (allele-level) resolution (hla_typing · Whole blood or buccal swab; no tumor tissue consumed · 1-2 weeks)

Bundles with. Single blood draw shared with the germline HR gene panel

A positive result would open.

- HLA-restricted TCR-T and ImmTAC agents (TCR-T / bispecific TCR · trial only)

Next step. Order now

BRAF V600E mutation

Priority: medium **Scope:** Tumor-agnostic **Status:** Not measured

BRAF V600E is actionable across nearly all primary sites, colorectal excepted. It is a rarity in leiomyosarcoma, and the reason it belongs on the list is that any comprehensive panel reports codon 600 whether or not anyone thought to ask.

BRAF V600E is untested and the prior probability in uterine leiomyosarcoma is very low. No separate order is warranted: the codon is covered by the comprehensive panel already recommended for TMB, MSI and the HR genes. Non-V600E class II and III BRAF alterations, should any turn up, do not carry this indication.

Recommended assay. Tumor NGS panel covering BRAF codon 600 (NGS_panel · Archival FFPE · 2-3 weeks)

Bundles with. Comprehensive DNA plus RNA genomic profiling on archival tissue

A positive result would open.

- dabrafenib plus trametinib (BRAF inhibitor plus MEK inhibitor · tumor-agnostic approval · NCT02034110)

Next step. Bundle with planned testing

References. nct:NCT02034110

HER2 protein overexpression (IHC 3+)

Priority: medium **Scope:** Tumor-agnostic **Status:** Not measured

HER2 IHC 3+ supports trastuzumab deruxtecan irrespective of primary site. Strong HER2 staining is uncommon in leiomyosarcoma, but nothing on file addresses it and the stain is routine.

HER2 protein expression is untested. Only 3+ carries the tumor-agnostic indication, and 3+ is unlikely in uterine leiomyosarcoma, so a 1+ or 2+ result should be reported as HER2-low and read as a weaker, less reliable signal rather than as a positive. A block is being cut for other stains anyway, which makes adding HER2 cheap and settles the question either way.

Recommended assay. HER2 IHC on tumor tissue (IHC · Archival FFPE block · 1 week)

Bundles with. Single archival block cut for the IHC set (MMR, PD-L1, HER2, B7-H3); ERBB2 copy number also comes off the comprehensive NGS panel

A positive result would open.

- trastuzumab deruxtecan (HER2-directed ADC · tumor-agnostic approval · NCT04482309)

Next step. Bundle with planned testing

References. nct:NCT04482309

RET gene fusion

Priority: medium **Scope:** Tumor-agnostic **Status:** Not measured

The RET fusion indication ignores primary site. RET fusions in uterine leiomyosarcoma are rare to the point of being anecdotal, but the RNA fusion assay ordered for NTRK answers RET in the same run.

RET fusion is untested and will almost certainly be negative in this histology. It earns a row because it costs nothing extra once an RNA fusion panel is ordered for NTRK, and because a positive would place an oral targeted agent ahead of further cytotoxic chemotherapy. A RET point mutation is a different finding and does not carry this indication.

Recommended assay. RNA-based fusion NGS panel covering RET (NGS_panel · Archival FFPE · 2-3 weeks)

Bundles with. Comprehensive DNA plus RNA genomic profiling on archival tissue (same order as NTRK)

A positive result would open.

- selpercatinib (selective RET inhibitor · tumor-agnostic approval · NCT03157128)

Next step. Bundle with planned testing

References. nct:NCT03157128

Fibroblast activation protein alpha (FAP)

Priority: medium **Scope:** Relevant across a tumor subset **Status:** Not measured

FAP is the panel's stromal target, and mesenchymal tumors carry it on the malignant cells as well as on the associated fibroblasts, which is why sarcoma shows up so reliably on FAPI PET. The panel's expression evidence is drawn from carcinoma stroma, so treat sarcoma relevance as well-supported by imaging rather than by an IHC series in this histology.

FAP is untested, and no FAP-directed agent is approved anywhere, so a positive result would only matter if a radioligand or bispecific trial were within reach. FAP IHC is not a routine clinical stain and most laboratories do not offer it, which makes speculative testing a poor use of a block. Revisit if a FAP-directed trial becomes a live option, at which point FAPI PET is usually the practical screening route and spends no tissue.

Recommended assay. FAP IHC on archival tissue, or FAPI PET where a FAP-directed program requires it (IHC · Archival FFPE · 1-2 weeks)

Bundles with. Archival IHC block if a FAP-directed program is in play; otherwise the trial's own screening assay

A positive result would open.

- ¹⁷⁷Lu-FAP-2286 (radioligand therapy · trial only)

- RO7300490 (bispecific (FAP-4-1BBL) · trial only)
- simlukafusp alfa (FAP-IL2v immunocytokine · trial only)

Next step. Defer

References. pmid:32601975

PD-L1 protein expression

Priority: medium **Scope:** Relevant across a tumor subset **Status:** Not measured

PD-L1 gates no approval in soft tissue sarcoma, so this is a supporting rather than a decisive result. It stays in scope because the pulmonary lesions were read as infiltrative and possibly inflammatory, which makes the immune phenotype of this tumor a question someone will raise.

PD-L1 has not been stained. In sarcoma it neither opens nor closes a labelled indication, so treat the result as supporting information for trial screening and for weighing any checkpoint-blockade proposal, not as a decision point in its own right. If it is stained, use the lung metastasis and match the clone and cutoff to whichever agent is under consideration, because a result generated for one companion diagnostic does not transfer to another indication.

Recommended assay. PD-L1 IHC using the clone and scoring system matched to the intended agent (IHC · Archival FFPE; the 2026 lung core is the more informative specimen for current immune phenotype · 1 week)

Bundles with. Single archival block cut for the IHC set (MMR, PD-L1, HER2, B7-H3)

A positive result would open.

- PD-1 / PD-L1 inhibitors (checkpoint inhibitor · trial only)

Next step. Bundle with planned testing

MAGE-A4 (HLA class I-presented cancer/testis antigen)

Priority: medium **Scope:** Relevant across a tumor subset **Status:** Not assessable yet

MAGE-A4 is the antigen behind the first approved TCR-T therapy for a solid tumor, and that approval is written for synovial sarcoma rather than sarcoma in general. Expression in leiomyosarcoma runs well below the 70-80% seen in synovial sarcoma, and neither the antigen nor the HLA type has been tested here.

MAGE-A4 status is unknown and stays unknown until the HLA type is back, since the whole class is allele-restricted. Even a positive stain would not make her eligible for afami-cel, which is licensed for synovial sarcoma alone, so the realistic route is a MAGE-A4 solid-tumor TCR-T trial. Include it in the same antigen stain set as PRAME rather than ordering it separately.

Recommended assay. MAGE-A4 IHC on archival tissue, contingent on HLA class I genotyping first (IHC · Archival FFPE · 1-2 weeks after HLA result)

Bundles with. Cancer/testis antigen IHC set (PRAME, MAGE-A4, NY-ESO-1) on one archival block, ordered only if HLA typing returns a restricting allele

A positive result would open.

- afamitresgene autoleucel (TCR-T · approved, other tumor type)
- uzatresgene autoleucel (TCR-T · trial only)

Next step. Reflex if positive

References. pmid:38554725; pmid:36624315

PRAME (HLA class I-presented cancer/testis antigen)

Priority: medium **Scope:** Relevant across a tumor subset **Status:** Not assessable yet

PRAME is expressed in a subset of sarcomas including leiomyosarcoma, and IMA203 TCR-T enrolls PRAME-positive HLA-A*02:01 solid tumors. Neither the antigen nor the restricting HLA allele has been tested, so this target cannot be called in either direction.

PRAME cannot be scored as present or absent, because neither the antigen nor the restricting HLA allele has been tested. Order HLA typing first: if she carries HLA-A*02:01, PRAME IHC is worth doing because it gates entry to the ImmTAC and TCR-T programs, all of which are trials rather than approved options. If she does not carry a restricting allele, this row closes without any tissue being spent.

Recommended assay. PRAME IHC on archival tissue, contingent on HLA class I genotyping first (IHC · Archival FFPE · 1-2 weeks after HLA result)

Bundles with. Cancer/testis antigen IHC set (PRAME, MAGE-A4, NY-ESO-1) on one archival block, ordered only if HLA typing returns a restricting allele

A positive result would open.

- brenetafusp (ImmTAC bispecific TCR · trial only)
- IMA203 (TCR-T · trial only)

Next step. Reflex if positive

AXL receptor tyrosine kinase surface expression

Priority: low **Scope:** Relevant across a tumor subset **Status:** Not measured

AXL marks mesenchymal and EMT-high tumor cells, which is the constitutive state of a sarcoma rather than an acquired one, and it tracks with chemotherapy resistance. The panel's expression evidence comes from carcinomas, so read this as plausible in leiomyosarcoma rather than established.

AXL has not been assessed, and no AXL-directed agent is approved: enapotamab vedotin and tilvestamab both stopped at completed early-phase studies. Testing would not change anything she can access now or at progression. Revisit only if an AXL-directed program opens with an expression-based entry criterion.

Recommended assay. AXL IHC on tumor tissue (IHC · Archival FFPE · 1-2 weeks)

Bundles with. Would ride on the archival IHC block if an AXL-directed program required documented expression

A positive result would open.

- enapotamab vedotin (ADC · trial only)
- tilvestamab (mAb · trial only)

Next step. Defer

NY-ESO-1 (CTAG1B, HLA class I-presented cancer/testis antigen)

Priority: low **Scope:** Relevant across a tumor subset **Status:** Not assessable yet

NY-ESO-1 is the cancer/testis antigen of synovial sarcoma and myxoid liposarcoma, where expression runs 70-80% and roughly 30%. Leiomyosarcoma is not one of those histologies, and no HLA type is on file either, so the target is unassessable rather than negative.

NY-ESO-1 cannot be called without an HLA type, and the prior probability in leiomyosarcoma is low. Lete-cel is being developed in synovial sarcoma and myxoid/round cell liposarcoma, so even a positive stain would not point at a trial written for this histology. Add it only if the cancer/testis antigen block is already being stained for PRAME and MAGE-A4.

Recommended assay. NY-ESO-1 IHC or RT-PCR, contingent on HLA class I genotyping first (IHC · Archival FFPE · 1-2 weeks after HLA result)

Bundles with. Cancer/testis antigen IHC set (PRAME, MAGE-A4, NY-ESO-1) on one archival block

A positive result would open.

- letetresgene autoleucel (TCR-T · trial only · NCT03967223)

Next step. Reflex if positive

References. nct:NCT03967223

Panel targets out of scope for this tumor

66 of 72 panel targets were screened and set aside as having no plausible connection to this patient's tumor type or stated features: ALPPL2 · ANTXR1 · CA9 · CD19 · CD200 · CD37 · CD46 · CD70 · CDCP1 · CDH17 · CDH3 · CDH6 · CEACAM5 · CLDN18 · CLEC12A · CT83 · DLK1 · DLL3 · EGFR · EGFR (variant III) · ENPP3 · EPCAM · EPHA2 · EPHB4 · ERBB2 · ERBB3 · F3 · FOLH1 · FOLR1 · FZD7 · GPC3 · GPRC5D · GUCY2C · HAVCR1 · IL13RA2 · IL3RA · ITGA3 · ITGB6 · L1CAM · LGR5 · LILRB4 · LY6E · LYPD3 · MAGEA1 · MAGEC2 · MERTK · MET · MMP14 · MSLN · MUC1 · NECTIN4 · PODXL · PSCA · PTK7 · ROR1 · SEZ6 · SLC34A2 · SLC44A4 · SLITRK6 · SSTR2 · SSX2 · STEAP1 · TACSTD2 · TNFRSF10B · TNFRSF17 · TSPAN8